Flight Training Lesson Guide
UPRT/EET

Revisions

Any questions about the content or use of this manual can be directed to:

Fleet Captain - Atlas Air Training Center or;
Standardsreport747@atlasair.com

Revision Highlights

- Added - to initiate a climb or descent to the Airspeed Unreliable.
- Guidance for number of takeoffs and Landings.

Lesson Outline

The lesson provides training and qualification in UPRT/EET procedures and bounced landing recovery.

- Stall prevention and recovery in clean configuration, takeoff and maneuvering configuration, and landing configuration.
- Manually controlled instrument departure and arrival.
- Manually controlled slow flight.
- Upset recovery maneuvers.
- Recovery from stall.
- Manually controlled loss of reliable airspeed.
- Recovery from bounced landing.

Grading

Demonstrate proficiency in UPRT/EET and bounced landing recovery training.

Document UPRT/EET and bounced landing certification on a PC form.

Document 3 Takeoffs and Landings on the grade sheet.

Objective

To train the skills and knowledge required to accomplish the maneuvers, procedures, and duties as captain or first officer.

Training

- Portable Electronic Flight Bag (pEFB).
- Glare-shield Checklists.
- QRH.

Briefing

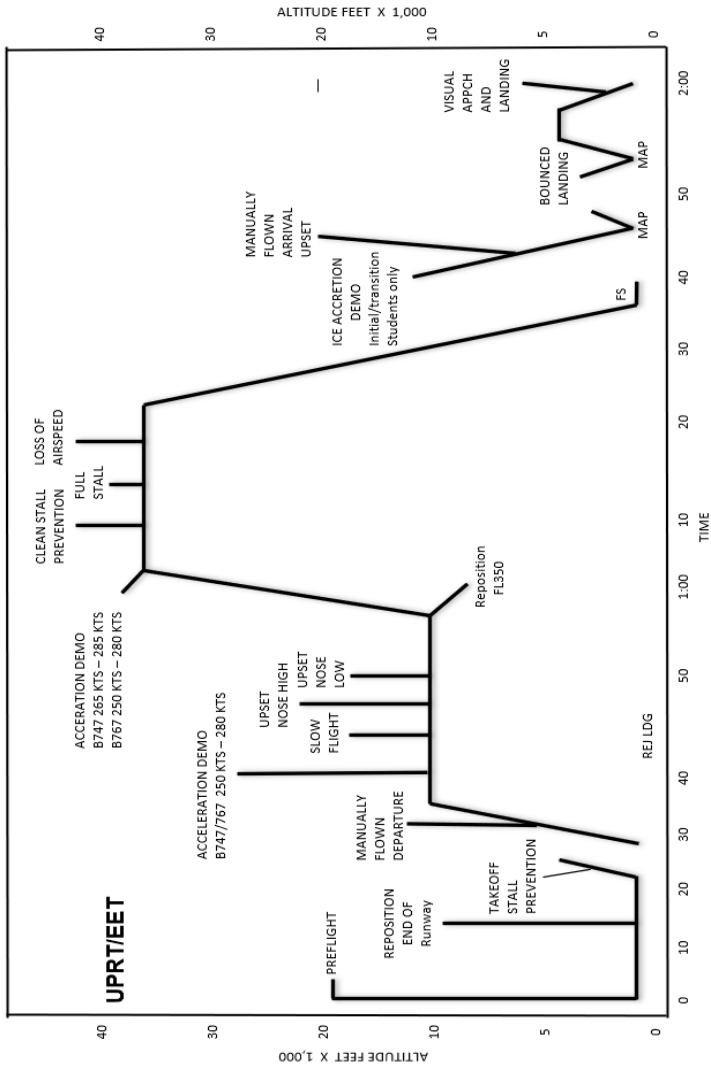
The briefing will be 2 hours in duration and consist of reviewing the student version of the UPRT/EET PowerPoint briefing. This will constitute the required UPRT/EET ground training.

- §121.423 – Pilot Extended Envelope Training
  - Extended envelope training must include.
    - Stall prevention and recover in clean configuration, takeoff and maneuvering configuration, and landing configuration.
    - Upset prevention and recovery.
    - Manually controlled slow flight.
    - Manually controlled loss of reliable airspeed.
    - Manually controlled instrument departure and arrival.
    - Upset recovery maneuvers.
    - Recovery from bounced landing.
  - Level C, or higher, simulator.
  - All maneuvers required every 2 years.

Simulator Presets

Simulator preset pictorials are representative and may not be identical to a specific simulator.

UPRT/EET and bounced landing Lesson Profile



OFP DATA –

ORIGIN: NEW YORK, (KJFK)
DESTINATION: NEW YORK, (KJFK)
FLIGHT NUMBER: GTI 400
ROUTE: KJFK SKORR_YANKEE TRANSITION
RUNWAY: 31L
DEPARTURE: SKORR_

RAMP FUEL…….. 56.4 **TAXI FUEL (INFO ONLY)**..... 0.9
RESERVE……..... 8.9 **COST INDEX**…….. 28
FLIGHT LEVEL…….. 370 **DISTANCE**……..……4401
DDG: NONE

LOAD SHEET DATA KILOS (LBS)

ZFW……...240.0 **TOGW**……296.4
CG……..21%

WEATHER

Kennedy International Information *"Alpha" XXxxz, Wind 240/08, Visibility 1 sm FG, sky condition Overcast 500, Temperature 23, Dew Point 22, Altimeter 29.92, Departing Runways 31L. On initial contact advise you have information Alpha"*

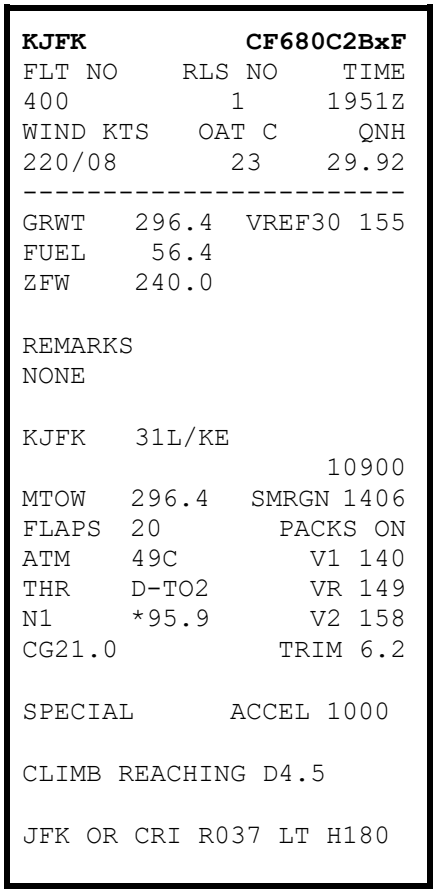
CLEARANCE

Clearance*, Cleared To Amsterdam Schipol airport, SKORR_ Departure, Radar Vectors, BOS. Climb and maintain 5,000 expect FL370 one zero minutes after departure, squawk 1747. Expect Runway 31L for departure.*

TLR DATA STRIP Kg / N1 (B5)

---DRY RWY – PTOW – ENG A/I OFF CALM WIND---
–

RWY	MTOW	MT	N1	CONFIG	FL	V1	VR	V2
31L	2964	49	95.9	D-TO2 PACKS ON	10	139	148	157



```
KJFK              CF680C2BxF
FLT NO     RLS NO     TIME
400             1     1951Z
WIND KTS    OAT C      QNH
220/08         23    29.92
-------------------------
GRWT    296.4  VREF30 155
FUEL     56.4
ZFW     240.0

REMARKS
NONE

KJFK    31L/KE
                      10900
MTOW    296.4   SMRGN 1406
FLAPS   20        PACKS ON
ATM     49C         V1 140
THR     D-TO2       VR 149
N1      *95.9       V2 158
CG21.0            TRIM 6.2

SPECIAL       ACCEL 1000

CLIMB REACHING D4.5

JFK OR CRI R037 LT H180
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*N1 may differ slightly due to differences between B1/B5 engines.

FFS UPRT/EET

Preflight

- Abbreviated procedures.
- Program FMC, set flaps, trim, airspeeds and MCP.
- Instructor will configure aircraft for takeoff on the runway.

Takeoff – Normal Procedures

- Instrument, Crosswind.
- Manually flown SID (Autopilot and Autothrottle off).
  - Student must manually set Takeoff thrust and Climb Thrust.
  - Review supplemental procedure – Manual Thrust Lever Takeoff.
- **Takeoff and maneuvering stall prevention and recovery training**
  - The exercise is designed to demonstrate how to prevent a takeoff stall, how to recognize and how to recover from it.
  - The pilot-in-training is required to recognize an approach to stall and recover. The pilot-in-training will confirm the autopilot and autothrottle are disengaged. The pilot-in-training is required to recover to stabilized flight with a pitch, thrust, and airplane configuration that corresponds to the desired airspeed.
  - Review the Atlas Stall Training supplement

Instructors Note:

Sim #3:

- Tablet password: uprt.
- If the V-N diagram fails to plot data points after a maneuver, from the maneuver page, close the application and restart.

ACCELERATION DEMONSTRATION

Objective:

Recognize how altitude affects thrust available.

General

The maneuver is conducted at low altitude and high altitude noting the difference in time it takes to accelerate 30 knots.

Initial Conditions (Low altitude)

- Altitude: 10000 ft.
- Configuration: Clean
- Airspeed: 250 kts
- Autopilot: Engaged.
- Autothrottle: Engaged.

Initial Conditions (High altitude)

- Altitude: FL 350.
- Configuration: Clean
- Airspeed: 265 kts
- Autopilot: Engaged.
- Autothrottle: Engaged.

1. Low/High Altitude Demo - Ask the student to estimate how long it takes to accelerate 30 knots?
2. Have student set speed to 250 kts (low altitude), 265 kts (High Altitude), allow the autothrottles to stabilize the speed.
3. Disengage the autothrottes, push the thrust levers full forward and note time it takes accelerate 30 kts.

   (Low – 250 kts – 280 kts)

   (High – 265 kts – 295 kts)
4. Note time difference between low and high altitude.
5. Lesson – Thrust doesn't does not affect recovery at high altitude. Pitch reduction is primary.

MANUALLY CONTROLLED SLOW FLIGHT

Objective:

Recognize the low energy or high drag configuration and the slow response to flight control and thrust inputs to enhance the pilot's knowledge of the low-speed handling qualities prior to stall training.

Manually controlled slow flight: The target speed for slow flight is VREF/-5, as calculated, for the configuration. Slow flight exposes the pilot to flight just above the stall speed and how to maneuver the airplane at this speed, in situations other than an approach, without stalling. The purpose is to reinforce the basic stall characteristics learned in academics and allow the pilot to obtain handling experience and motion sensations when operating the airplane at slow speeds in the impending regime in various airplane attitudes, configurations, bank angles, and altitudes, including high altitudes.

The maneuver will be conducted in the following two scenarios:

Initial Conditions (Low altitude)

- Altitude: 10000 ft.
- Maximum landing weight.
- Configuration: Transition from clean to Landing flaps.
- Airspeed: Vref-5
- Autopilot: Disengaged.
- Autothrottle: Disengaged.
- Flight Director – Off.

1. Have the pilot-in-training maintain altitude while configuring to the landing configuration.

   Encountering stick shaker without executing a stall recovery could lead to negative training.

2. Once student is in the landing configuration and at Vref -5:
   a. Turn left and right, and change direction of turn, to observe changing handling characteristics.
   b. Turns through 90º left and right, at bank angles appropriate to speed and configuration.
   c. Climb and descend at 500 feet per minute (fpm) while in a turn.

d. Recover to appropriate airspeed for the configuration and establish the appropriate altitude and heading.
e. Recovery is complete when straight and level un-accelerated flight is achieved.
f. One complete ask crew to accelerate to 250 knots and retract flaps to UP.

Common Pilot-in-Training Errors:

a. Inadequate back-elevator pressure as power is reduced, resulting in altitude loss.
b. Excessive back-elevator pressure as power is reduced, resulting in a climb, followed by a rapid reduction in airspeed and loss of flight control effectiveness.
c. Inadequate compensation for adverse yaw during turns.
d. Fixation on the airspeed indicator.
e. Failure to anticipate changes in lift as flaps are extended or retracted.
f. Inadequate power management.
g. Inability to adequately divide attention between airplane control and orientation.

UPSET
NOSE-HIGH

Objective:

Develop skills for recovery from a nose-high airplane upset.

General Description

This exercise should be used for initial training. The pilot is exposed to airplane nose-high aerodynamic characteristics. The exercise is designed to allow the pilot-in-training to develop proficiency in techniques for recovering from a nose-high airplane upset. Specifically, the pilot-in-training is required to recover from a minimum of a 40-deg, nose-high upset by recognizing and confirming the situation, verifying that the autopilot and autothrottle are disengaged, and applying appropriate recovery techniques. The first iteration requires the pilot-in-training to use up to full nose-down elevator. The second iteration requires the pilot-in-training to roll the airplane as a technique for reducing the pitch. The third iteration requires the pilot-in-training to use thrust reduction as a pitch-reduction recovery technique, if the airplane model has underwing-mounted engines. All iterations require the pilot to complete the recovery by rolling to wings level, if necessary, and, at the appropriate time, checking airspeed and establishing a final recovery pitch attitude.

SELECT <u>ONE</u> NOSE HIGH UPSET EXCERSISE:

Exercise 1 - Use of nose down Elevator

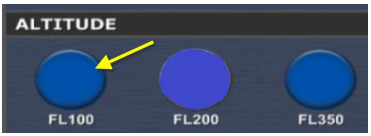
Exercise 2 - Use of bank angle

Exercise 3 - Use of thrust reduction

Upset Exercise 1 – Use of Nose Down Elevator

1. Initiate the exercise by selecting the following presets:





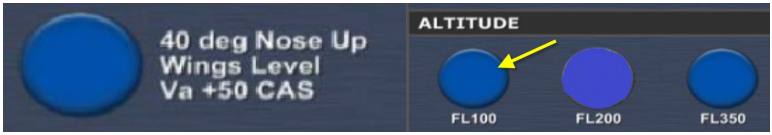
2. Select EXECUTE.
3. Prior to releasing FLIGHT FREEZE discuss the following with the student to initiate recovery:
    - Startle effect – Instructors must realize, and note to the student, that there will be a time delay between the event and the application of the recovery maneuver during line operations. The student will begin the maneuver with their hands on their lap (rather than holding the control yoke and thrust levers - prepared) before the instructor releases freeze. This will simulate an approximate startle effect.
    - Have the pilot-in-training note the pitch attitude for the initial conditions.
    - Verify that the autopilot and autothrottle are disengaged.
    - Counter pitch by use of up to full nose down elevator and, if required, by using stabilizer trim to relieve elevator control pressure. A steady nose down pitch rate should be achieved and it should be noted that the airplane would be near zero g and the associated characteristics of such.
    - Excessive use of pitch trim or rudder may aggravate the upset situation or may result in high structural loads.
    - The recovery is completed when approaching the horizon by checking airspeed, adjusting thrust, and establishing the appropriate pitch attitude and stabilizer trim setting for level flight.
4. Check the Alpha Beta plot. If the validated FSTD envelope was exceeded (airplane exceeds limits of the simulator, data is invalid), advise the pilot, provide instruction and repeat the scenario to prevent negative training.
5. Instructors will review the V-N diagram with the student to ensure that recovery was made within the aircraft limits.

Common Pilot-in-Training Errors:

a. Fails to simulate disengaging the autopilot and autothrotle
b. Hesitates to use up to full control input.
c. Over trims nose down stabilizer.
d. Fails to induce nose up stabilizer trim.

Upset Exercise 2 - Use of Bank Angle

1. Initiate the exercise by selecting the following presets:



2. Select EXECUTE.
3. Prior to releasing FLIGHT FREEZE discuss the following with the student to initiate recovery:

    - Startle effect – Instructors must realize, and note to the student, that there will be a time delay between the event and the application of the recovery maneuver during line operations. The student will begin the maneuver with their hands on their lap (rather than holding the control yoke and thrust levers - prepared) before the instructor releases freeze. This will simulate an approximate startle effect.
    - Have the pilot-in-training note the pitch attitude for the initial conditions.
    - Verify that the autopilot and autothrottle are disengaged.
    - Roll the airplane 30 to 60 degrees of bank until a nose down pitch rate is achieved. Some forward control pressure may be required. Avoid bank angles more than 60 deg.
    - Excessive use of pitch trim or rudder may aggravate the upset situation or may result in high structural loads.
    - The recovery is completed when approaching the horizon by checking airspeed, adjusting thrust, and establishing the appropriate pitch attitude and stabilizer trim setting for level flight.

4. Check the Alpha Beta plot. If the validated FSTD envelope was exceeded (airplane exceeds limits of the simulator, data is invalid), advise the pilot, provide instruction and repeat the scenario to prevent negative training.
5. Instructors will review the V-N diagram with the student to ensure recovery was made within aircraft limits.

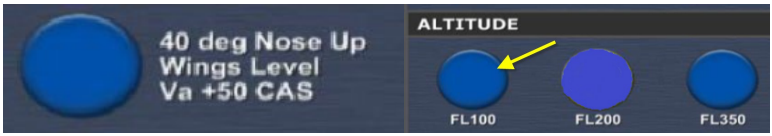
Common Pilot-in-Training Errors

a. Achieves the required roll too slowly, which allows the nose to drop too slowly and airspeed to become excessively low.

b. Continues the roll past what is required to achieve a nose down pitch rate; therefore, the difficulty of recovery is unnecessarily increased.
c. Rolls out at a pitch attitude that is too high for conditions and encounters an approach to stall.

Upset Exercise 3 – Use of Thrust Reduction

1. Initiate the exercise by selecting the following presets:



2. Select EXECUTE.
3. Prior to releasing FLIGHT FREEZE discuss the following with the student to initiate recovery:
    - Startle effect – Instructors must realize, and note to the student, that there will be a time delay between the event and the application of the recovery maneuver during line operations. The student will begin the maneuver with their hands on their lap (rather than holding the control yoke and thrust levers - prepared) before the instructor releases freeze. This will simulate an approximate startle effect.
    - Have the pilot-in-training note the pitch attitude for the initial conditions.
    - Verify that the autopilot and autothrottle are disengaged.
    - Counter pitch by use by reducing thrust to approximately midrange until a detectable nose down pitch rate is achieved.
    - Excessive use of pitch trim or rudder may aggravate the upset situation or may result in high structural loads.
    - The recovery is completed when approaching the horizon by checking airspeed, adjusting thrust, and establishing the appropriate pitch attitude and stabilizer trim setting for level flight.
4. Check the Alpha Beta plot. If the validated FSTD envelope was exceeded (airplane exceeds limits of the simulator, data is invalid), advise the pilot, provide instruction and repeat the scenario to prevent negative training.
5. Instructors will review the V-N diagram with the student to ensure that recovery was made within aircraft limits.

Common Pilot-in-Training Errors

a. Fails to simulate disengaging the autopilot and autothrottle.
b. Fails to reduce thrust sufficiently to obtain nose down pitch.

a. Reduces thrust excessively.

UPSET

NOSE-LOW

Objectives:

Demonstrate low-speed and high-speed accelerated stalls.

Develop skills for recovery from a nose-low airplane upset.

General Description

The exercise is designed to demonstrate what an approach to accelerated stall is and how to recover from it. The pilot-in-training is required to recover from a minimum of a 20 deg nose-low upset. High-bank-angle (up to inverted flight), nose-low upset iterations are used. To recover, the pilot-in-training recognizes and confirms the situation and verifies that the autopilot and autothrottle are disengaged. Thrust is adjusted for the appropriate energy condition. For a satisfactory nose-low recovery, the pilot-in-training must avoid ground impact and accelerated stall and respect g-force and airspeed limitations. The pilot-in-training is required to recover to stabilized flight with a pitch, thrust, and airplane configuration that corresponds to the desired airspeed.

SELECT ONE NOSE LOW UPSET EXCERSISE:

Exercise 1 – Nose low recovery in a bank

Exercise 2 – High bank angle/inverted flight

Upset Exercise 1: - Nose Low Recovery in a Bank

1. Initiate the exercise by selecting the following presets:



2. Select EXECUTE.
3. Ensure IMC.
4. Prior to releasing FLIGHT FREEZE discuss the following with the student to initiate recovery:

   - Startle effect – Instructors must realize, and note to the student, that there will be a time delay between the event and the application of the recovery maneuver during line operations. The student will begin the maneuver with their hands on their lap (rather than hold the control yoke and thrust levers - prepared) before the instructor releases freeze. This will simulate an approximate startle effect.
   - Have students practice recovery with and without the speedbrakes.
   - Have the pilot-in-training note the pitch and bank attitude for the initial conditions.
   - Verify that the autopilot and autothrottle are disengaged.
   - Roll in the shortest direction to wings level. It is important that positive G force not be increased or that nose-up elevator or stabilizer trim be used until the airplane approaches wings level.
   - Then apply nose up elevator; applying stabilizer trim, if necessary.
   - Excessive use of pitch trim or rudder may aggravate the upset situation or may result in high structural loads.
   - Adjust thrust and drag (speedbrakes) as necessary.
   - The recovery is completed when approaching the horizon by checking airspeed, adjusting thrust, and establishing the

appropriate pitch attitude and stabilizer trim setting for level flight.

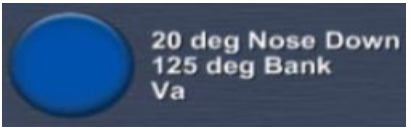
5. Check the Alpha Beta plot. If the validated FSTD envelope was exceeded (airplane exceeds limits of the simulator, data is invalid), advise the pilot, provide instruction and repeat the scenario to prevent negative training.
6. Review the V-N diagram with the student to ensure that recovery was made within aircraft limits.

Common Pilot-in-Training Errors

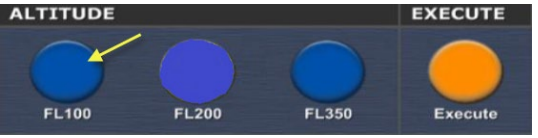
a. Forgets to disengage the autopilot and or autothrottle.
b. Fails to use full control inputs.
c. Initiates pull-up before approaching wings level.
d. Attempts to precisely obtain wings level and delays pull-up.
e. Enters secondary stall.
f. Exceeds positive g force during pull-up.
g. Fails to reduce thrust to idle for high speed.
h. Fails to use speedbrakes, if required.
i. Achieves inadequate pull-up to avoid ground impact.

Upset Exercise 2 - High Bank Angle/Inverted Flight

1. Initiate the exercise by selecting the following presets:







2. Select EXECUTE.
3. Ensure IMC
4. Prior to releasing FLIGHT FREEZE discuss the following with the student to initiate recovery:

    - Startle effect – Instructors must realize, and note to the student, that there will be a time delay between the event and the application of the recovery maneuver during line operations. The student will begin the maneuver with their hands on their lap (rather than hold the control yoke and thrust levers - prepared) before the instructor releases freeze. This will simulate an approximate startle effect.
    - Have students practice recovery with and without the speed brakes.
    - Have the pilot-in-training note the pitch and bank attitude for the initial conditions. Noting the reduced ability to visually detect the horizon.
    - Verify that the autopilot and autothrottle are disengaged.
    - Roll in the shortest direction to wings level. It is important that positive G force not be increased or that nose-up elevator or stabilizer trim be used until the airplane approaches wings level.
    - Then apply nose up elevator, applying stabilizer trim, if necessary.
    - Excessive use of pitch trim or rudder may aggravate the upset situation or may result in high structural loads.
    - Adjust thrust and drag (speedbrakes) as necessary.
    - The recovery is completed when approaching the horizon by checking airspeed, adjusting thrust, and establishing the

appropriate pitch attitude and stabilizer trim setting for level flight.

5. Check the Alpha Beta plot. If the validated FSTD envelope was exceeded (airplane exceeds limits of the simulator, data is invalid), advise the pilot, provide instruction and repeat the scenario to prevent negative training.
6. Instructor will review the V-N diagram with the student to ensure recovery was made within the aircraft limits. See note.

Note: It may be difficult for the student to keep the airspeed within the V-N diagram. It is more important that the student remain within the “G” envelope.

Common Pilot-in-Training Errors

a. Forgets to disengage the autopilot or autothrottle.
b. Fails to unload.
c. Fails to use sufficient control inputs.
d. Initiates pull-up before approaching wings level.
e. Attempts to precisely obtain wings level and delays pull-up.
f. Exceeds positive g-force limits during pull-up.
g. Fails to reduce thrust to idle for high speed.
h. Fails to use speed brakes, if required.
i. Achieves inadequate pull-up to avoid ground impact.

STALL

HIGH ALTITUDE

Objectives:

Develop skills for recovery from a high-altitude stall.

General Description

The exercise is designed to demonstrate how to prevent a high-altitude stall, how to recognize and how to recover from it.

The pilot-in-training is required to recognize an approach to stall and recover. The pilot-in-training will also be required to recognize a full aerodynamic stall and recover. The pilot-in-training will confirm the autopilot and autothrottle are disengaged. The pilot-in-training is required to recover to stabilized flight with a pitch, thrust, and airplane configuration that corresponds to the desired airspeed.

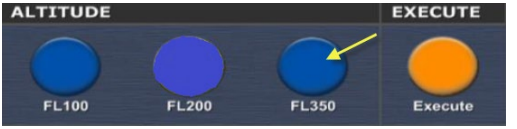
Approach to Clean Stall Training

- Review the Atlas Stall Training supplement

Full Aerodynamic Stall Exercise

1. Initiate the exercise by selecting the following presets:





2. Select EXECUTE.
3. Autopilot and autothrottle: Engaged, HDG SEL and VNAV.
4. Prior to releasing FLIGHT FREEZE discuss the following with the student to initiate the stall and recovery:

5. Startle effect – Instructors must realize, and note to the student, that there will be a time delay between the event and the application of the recovery maneuver during line operations. The student will begin the maneuver with their hands on their lap (rather than hold the control yoke and thrust levers - prepared) before the instructor releases freeze. This will simulate an approximate startle effect.

- Autopilot remains engaged.
- Thrust levers to idle.
- Allow the aircraft to bleed off airspeed.

- Allow the aircraft to slow to a fully developed stall as indicated by a negative rate of descent indication on the IVSI while increasing pitch (maintaining altitude is not required)

6. Once a negative rate of descent is indicated the student should recover.
    - The pilot will perform a deliberate and smooth reduction of AOA.
    - Positive recovery from the stall event is paramount. There is no predetermined value for altitude loss and maintaining altitude during recovery is not required.
    - Appropriate application of thrust to accelerate and enable a positive recovery. (remembering the acceleration demo, pitch is primary first, thrust secondary)
    - Establishing the appropriate AOA takes precedence over roll control (attempting to maintain wings level) for positive recovery from the stall event.
    - Intermittent secondary stall warnings, but not secondary stalls, may be acceptable due to the associated recovery challenges at altitude because of the lack of aerodynamic damping.
    - The maneuver is considered complete once a safe speed is achieved and the airplane stabilized.
7. Check the Alpha Beta plot. If the validated FSTD envelope was exceeded (airplane exceeds limits of the simulator, data is invalid), advise the pilot, provide instruction and repeat the scenario to prevent negative training.
8. Instructors will review the V-N diagram with the student to ensure that recovery was made within aircraft limits.

Common Pilot-in-Training Errors

a. Not disconnecting the autopilot and/or autothrottle prior to reducing AOA.
b. Recovery is attempted with thrust instead of reducing AOA.
c. Not maintaining a nose-down input until the impending stall cues are eliminated.
d. Insufficient pitch down to allow desired energy conversion of altitude to airspeed.
e. Pilot fails to promptly recover from a secondary stall.
f. Reluctance to sacrifice significant altitude.
g. Pilot fails to distinguish between high-speed buffet and low-speed stall.

h. Pilot increases the load factor too quickly and gets multiple impending stalls.
i. Inappropriate use of rudder.
j. Pilot prioritizes roll control (attempting to level the wings) before reducing AOA.
k. Not disconnecting the autopilot and/or autothrottle/autothrust prior to reducing AOA.

LOSS OF RELIABLE AIRSPEED

Objectives:

The pilot will recognize the airspeed discrepancy, determine airspeed data is erroneous, and apply the appropriate non-normal procedure while maintaining airplane control using pitch and power targets.

General Description

The scenario will be conducted at or near the maximum operating altitude in instrument meteorological conditions (IMC). Use of flight simulation training device (FSTD) capabilities to induce erroneous airspeed indications may include:

- Full pitot/static blockage or icing.
- Air data computer failures.

Initial Conditions

- Altitude: FL350
- Airspeed: Mach.84
- Autopilot: Engaged.
- Autothrottle: Engaged.

1. Begin the exercise while in cruise flight.
2. Select Pitot tube blockage for Capt and FO (Ice & Rain menu).
3. Give a clearance to descend.
4. The exercise is complete after a visual approach and landing.
   (airspeed will decrease in a descent and increase in a climb).
5. The pilot recognizing the erroneous airspeed data indication will verbally announce the discrepancy.
6. The pilot flying will maintain control of the airplane and call for the appropriate checklist.
7. At the conclusion of the scenario, the instructor will discuss available airplane AOA indications.

Common Pilot-in-Training Errors

a. The importance of pitch control and AOA is not recognized.
b. Use of large thrust changes.
c. Failure to complete the appropriate non-normal checklist.
d. Over controlling the airplane, especially pitch.

ICE ACCRETION DEMONSTRATION

(INITIAL/TRANSITION TRAINING STUDENTS ONLY)

Objectives:

The pilot will recognize the effects of Ice Accretion to aircraft performance and engine vibration due to fan blade icing.

General Description

The scenario will be conducted at 10,000 feet at a set airspeed with the autothrottle off. Icing will be selected; the student will observe a speed reduction and engine vibration (noting engine vibration monitors on the engine synoptic) as ice begins to accumulate. Once at 100 % ice accretion as indicated by the instructor, the student will turn on the wing and nacelle heat and observe the effects of ice dissipating.

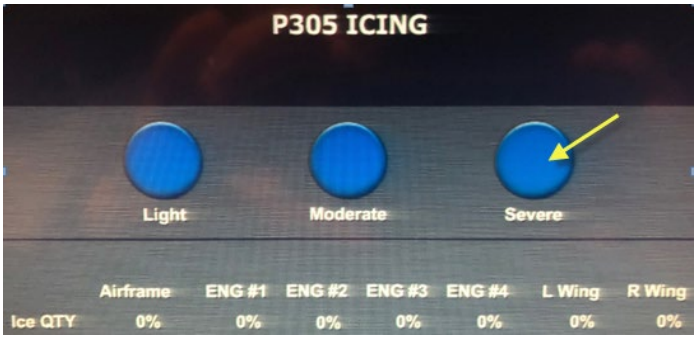
1. Initiate the exercise by selecting the following presets:





2. Set OAT:5 degrees
3. Set cloud cover above 10,000
4. Select EXECUTE.
5. Prior to releasing FLIGHT FREEZE ensure and discuss the following:
    - Autopilot: Engaged.
    - Autothrottle: Off.
    - Select engine synoptic.
    - Wing and Nacelle Ice: Off.
    - Discuss, once selected, ice will start to accumulate. As ice accumulates, engine vibrations start to increase. (note engine vibration monitors) and slight airspeed decrease.
    - Once 100% accretion is obtained (as noted by the instructor observing the IOS) Student will select wing and nacelle heat and note vibration reduction and airspeed increase.
6. Allow the aircraft to stabilize once Flight Freeze is released

7. Select Icing (severe)



8. Advise student when at 50% and 100% during ice accumulation (as noted on the IOS) and as the ice melts when wing and nacelle heat is activated. Advise when at 0% ice accumulation.
9. Demonstration complete.

MANUALLY FLOWN ARRIVAL WITH UPSET

Objectives:

The pilot will recognize an upset during an arrival caused by wake turbulence.

General Description

The pilot is exposed to airplane nose-high aerodynamic characteristics. The exercise is designed to allow the pilot-in-training to develop proficiency in techniques for recovering from an upset during an arrival.

Startle/Surprise

Because upsets that occur in normal flight operations are unplanned and inadvertent, pilots may be startled or surprised, adversely impacting recognition or recovery. Instructors need to plan scenarios to balance potential for startle or surprise while applying sound judgement with respect to realism and fidelity and respecting the capabilities and limitations of the FSTD. It is crucial for the instructor to adopt and foster a spirit of collaborative learning when inducing startle or surprise so as not to inappropriately attempt to trap a pilot or destroy confidence in the training session.

Exercise

1. Establish initial conditions:
    - On a published segment of a STAR.
    - Airspeed: Appropriate for configuration
    - Autopilot: Disengaged.
    - Autothrottle: Disengaged.

****THIS IS A SURPRISE - DO NOT ADVISE THE STUDENT PRIOR****

2. Initiate the exercise by selecting the following presets:



1. Select ACTIVATE - **(THIS IS A SURPRISE - DO NOT ADVISE THE STUDENT PRIOR)**
3. Verify the student confirms the situation, rolls to approaching wings level, then applying nose up elevator; applying stabilizer trim, if necessary; and adjusting thrust and drag as necessary.
4. Check the Alpha Beta plot. If the validated FSTD envelope was exceeded (airplane exceeds limits of the simulator, data is invalid), advise the pilot, provide instruction and repeat the scenario to prevent negative training.
5. Instructors will review the V-N diagram with the student to ensure that recovery was made within aircraft limits.

Common Pilot-in-Training Errors

a. Fails to use full control inputs.
b. Initiates pull-up before approaching wings level.
c. Attempts to precisely obtain wings level and delays pull-up.
d. Enters secondary stall.
e. Exceeds positive g force during pull-up.
f. Fails to reduce thrust to idle for high speed.
g. Fails to use speedbrakes, if required.
h. Achieves inadequate pull-up to avoid ground impact.

- **Landing configuration stall prevention and recovery training**
    - The exercise is designed to demonstrate how to prevent a landing configuration stall, how to recognize and how to recover from it.
    - The pilot-in-training is required to recognize an approach to stall and recover. The pilot-in-training will confirm the autopilot and autothrottle are disengaged. The pilot-in-training is required to recover to stabilized flight with a pitch, thrust, and airplane configuration that corresponds to the desired airspeed.
    - Review the Atlas Stall Training supplement

- **Bounced Landing Recovery**

Objectives:

Develop skills to recognize and recover from a bounced or skipped landing.

Exercise

1. If applicable, vector or reposition on a segment of the approach
   - Visual approach.
   - Autopilot may be used.
   - No autoland – landing must be manual.
   - Load ILS and approach speeds.
   - Configuration and speed appropriate for position.
2. Release FLIGHT FREEZE
3. Initiate the exercise by selecting the following presets:



4. Select ACTIVATE

If no automated preset

1. No automated presets - While in the flare the instructor will loudly announce “BOUNCED LANDING” when the GPWS calls out “20” feet.

When the bounced landing occurs

1. The pilot will execute a go-around by:
   a. Holding or re-establish a normal landing attitude.
   b. Apply go-around thrust and use normal go-around procedures.
   c. Give vectors back to the field for a visual approach and landing.

Recording Takeoffs and Landings.

- **Please ensure that the paperwork shows 3 takeoffs and landings.**